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Weightage 192424287 Assessment Tool 3: Class test 192424287
Course Name: Computer Vision with Opencv
Course Code: DSA02 Weightage: 10%

1. An image appears too bright. Explain and justify the most suitable grey level transformation to improve visibility, clearly describing the concept, application process and expected outcome

A. Use negative Transformation. It converts bright pixels into dark pixels using $s = L-1-r$, reducing excessive brightness and improving visibility

2. A dark image lacks detail. Describe and justify an appropriate grey level transformation to enhance visibility explaining how it improves low-intensity regions with proper reasoning

A. Use log transformation. It increases low-intensity pixel values, making dark regions brighter and revealing hidden details.

3. An image has poor contrast. Explain a suitable histogram-based technique for contrast enhancement and illustrate how it improves image quality with logical reasoning

A. Use histogram equalization. It redistributes pixel intensities to improve image contrast and makes objects more visible

4. An low-contrast image requires uniform intensity distribution. Describe and justify the method you would apply by explaining the concept working process and its effect on intensity levels

A Apply histogram equalization it spreads intensity values uniformly over the gray-level range, enhancing the overall contrast

5 An image shows uneven brightness across regions explain and apply an appropriate enhancement technique to correct non-uniform illumination providing a structured and accurate explanation

A Use Adaptive histogram equalization (CLAHE). it enhances contrast locally and corrects non-uniform illumination without over-enhancing the image

6. A noisy image contains random intensity variations explain and justify the most suitable smoothing filter, including its working mechanism and impact on noise reduction

A Use a median filter it replaces each pixel with the median of neighboring pixels, effectively removing salt and pepper noise while preserving edges

7 An image requires noise reduction without significant loss of details Describe and apply an appropriate filtering approach. explaining how it balances noise removal and detail Preservation

A. use a median filter it replaces each pixel with the median of neighboring pixels. effectively removing salt by considering both pixel distance and intensity differences

8 An image appears blurred after smoothing. Explain and justify a suitable sharpening method to restore details including its working principle and expected results

A. use the laplacian filter it enhances edges and restores image sharpness by emphasizing high--frequency components

9 fine details in an image are not clearly visible. Describe and justify the filtering technique used to enhance details. Explaining its affect on high -frequency components

A. use high- Pass filtering. it enhances high-frequency components such as edges and fine details making the image appear sharper.

10 A noisy image must be PreProcessed before further analysis. Explain and apply a suitable spatial filtering method, ensuring a clear, structured explanation with justification

A. Use a Gaussian filter. it smooths the image by reducing random noise, making it suitable for further image Processing tasks such as segmentation and edge detection